Notes on Learning

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1 Week 1

Human minds employ two different forms of thinking: focused and diffused modes.

- In focused mode, new thoughts tend to follow preestablished routes in the mind. In this mode, learning new concepts is difficult.
  - Repeating a problem you are confronted with allows for these routes to be solidified, kind of like ruts in a road.
- In diffused mode, new thoughts are free to touch a variety of concepts, and establish light new routes. This is where 'intuitive jumps' come from.

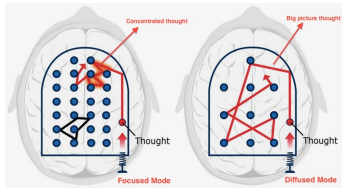

Figure 1: Focused and Diffused Thinking

- Along with this, are unconscious and conscious forms of processing thought. Conscious thought tends to be more on the focused end; it's targeted specifically toward one item.
- Unconscious thought, however, is far more diffused. It occurs when we aren't attempting specifically to think about any one thing; performing repetitive or menial activities often allows this to occur.

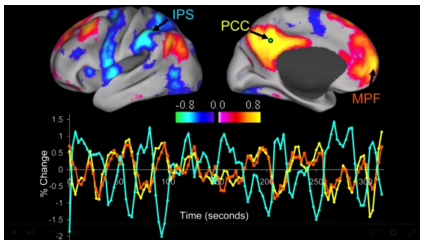

Figure 2: Conscious vs. Unconscious thought

Sleep allows new dendritic nodes to form on dendridic branches on neurons in the mind – meaning, sleep!
Procrastination: Unpleaseant ideas are linked to the pain centers of the brain, so by nature we seek nicer alternatives.

Introduction to Memory:

- Working memory: items that are being immediately processed.
  - Can hold about 4 'chunks' of information. Inefficient mental blackboard.
  - Think like this as RAM of a computer.
- Long term memory: items that are stored permanatly.
  - Think of a hard drive that needs to be written to several times to store a single piece of data; old data can also be overwritten.
  - Spaced repetition: extend over a period of time, with spaces in between.

Importance of sleep:

- Too little sleep can result in various disorders, particularly over a long period of time. Getting enough of it is critical to ensure ability to function nominally daily.
- Greater ability to deal with complex problems; allows for mind to better synthesize problems together.
- Dreaming about learned items also is a powerful method to retain info long-term.

Methods of Learning:

- Get into the thick of it – learning by doing.
- Learning a subject you are disinterested in is difficult, but using questions actively oft. helps disrupt the meniality.

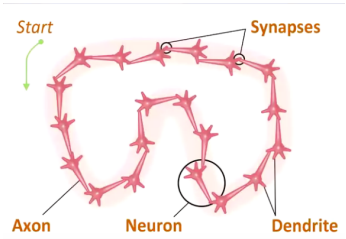

Figure 3: Graphic of Memory

- Diffuse mode – remembering these thoughts sometimes is difficult.
- Multitasking isn't necessarily a bad thing; but is more efficient to focus on a single item.
- All of the neurons you had in your brain now you have had since birth. Connections change and alter, and some die every once in a bit.
- New neurons are created in hippocampus throughout whole life, even past young adult stage.
- Social stimulation and active participation also influences neuron creation.
- Exercise also influences this rate of creation (and survival).
- Being in a creative environment potentially helps with enhancing creativity.
- Maintaining creativity in onslaught of repetitivity is difficult – consistently attempt to maintain excitement.
- Intelligence $\neq$ success. Passion and persistence are often more valuable traits.